

VCS Monitoring Report

Guangdong Nan Ao 26MW Wind Power Project

Monitoring period:
01/01/2008-14/12/2008

Version 2.0
26 August 2009
Climate Bridge Ltd.

1. Introduction

The purpose of this report is to calculate the emission reductions generated by Guangdong Nan Ao 26MW Wind Power Project¹ located in Nan Ao Island, Guangdong Province of China during the crediting and monitoring period covered by this report, and to serve as basis for the verification and issuance of corresponding VCU.

1.1 Monitoring Period

01/01/2008-14/12/2008

1.2 Document Reference

Version: 2.0

Date: 26 August 2009

2. General description of the project

2.1 Summary

The Guangdong Nan Ao 26MW Wind Power Project (hereafter refers to the project) involves construction and operation of a wind farm with a total capacity of 23.25 MW and a 110 KV substation at Nan Ao Island. The project construction commenced in April 2007, and the total 31 installed wind turbines (each with a capacity of 0.75 MW) started generating power in January 2008. The project utilizes local wind resources to generate electricity and supply power to the Guangdong Provincial Power Grid that is eventually connected to the Southern China Power Grid (SCPG).

The project was registered as a CDM project on 15 December 2008 (project reference number of 1963), and therefore, the project VCU crediting period started from 01/01/2008 through 14/12/2008. During the entire VCU crediting period, the total net electricity supplied to the grid from the project is about 52,311 MWh², displacing an equivalent amount of power generated from SCPG which is dominated by fossil fuel-fired power plant, and helping reduce greenhouse gas (GHG) emissions by some 48,524 tCO₂e.

The project is located in Nan Ao Island, Shantou City, Guangdong Province, People's Republic of China, and its geographical coordinates are 117°5'27.73"E and 23°25'12.63"N. Figures 2-1 shows the geographical location of the proposed project.

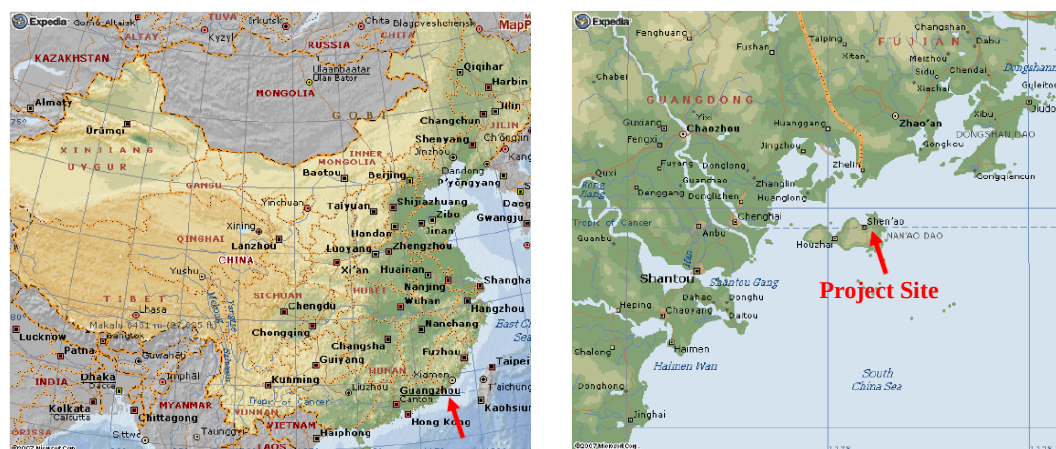


Figure 2-1 Location of proposed project

¹ UNFCCC project reference number is 1963

² Statistics from monthly power sales receipts in 2008

2.2 Technical description of the project activity

The project involves the installation of 31 wind turbines with each capacity of 750 kW. The wind turbines were supplied by Xinjiang Gold Wind Science and Technology Co., Ltd., and the selected model is S48/750 with pitch- and speed regulated. The annual operating hour of the project is estimated to be around 2,477h at full capacity. The main technical parameters of the wind turbine are provided in the following table 2-1.

The project also built a 110kV substation on Nan Ao Island, through which the project is connected to the 110kV Jinniu Transformer Substation, and the power generated is finally sold to China Southern Power Grid.

Table 2-1 Technical parameters of the wind turbine

Item	Unit	Index
Rated capacity	kW	750
Number of blades		3
Rotor diameter	M	44
Swept area	M2	2124
Startup speed	m/s	4
Rated wind speed	m/s	16
Shutoff speed	m/s	25
Rated voltage of generator	V	690

3. Monitoring methodology and plan

3.1 Monitoring methodology

The baseline and monitoring methodology adopted by the project is ACM0002 (version 6): “Consolidated baseline methodology for grid-connected electricity generation from renewable sources”

3.2 Data and parameters to be monitored

After the project started operation, another local wind farm company, Shenzhen Nanguo Energy Co., installed eight new turbines with each capacity of 750 kW next to the project site, and the earliest operation among the new turbines started on 5 October 2008. Please see table 3-1 below for details on operation date of eight turbines.

Table 3-1. Installation Date of Eight Turbines

Turbine No.	Operation Date
New turbine 1# (C7)	9 Oct. 2008
New turbine 2# (C8)	11 Nov. 2008
New turbine 3# (C9)	21 Oct. 2008
New turbine 4# (C10)	22 Oct. 2008
New turbine 5# (C11)	27 Oct. 2008
New turbine 6# (D9)	30 Oct. 2008
New turbine 7# (D10)	5 Oct. 2008
New turbine 8# (D11)	14 Oct. 2008

The eight new wind turbines share the 110 kv substation and the same ammeter with the proposed project. Therefore, during the period of 01/01/2008 till 04/10/2008, the total net electricity supplied to the grid by the proposed project activity is calculated using the total electricity supplied to the grid minus the total electricity imported from the grid. During

05/10/2008 till 14/12/2008, the total net electricity supplied to the grid by the proposed project activity is calculated by using the total electricity supplied to the grid minus both the total electricity imported from the grid and the total amount of electricity generated by the eight new wind turbines.

The total electricity supplied to and imported from the grid including the project activity and the 8 new turbines is monitored by the main meter installed at the substation. In order to exclude the power generation from the eight new turbines during the VCU crediting period, the electricity generated by the eight new turbines is also monitored by power meters installed inside of each wind turbine.

The model of the power meter for monitoring eight new turbines is Schneider Electric's PM810 power meter (63230-500-201A3). The power accuracy of the PM 810 under IEC61036 equals to China's Alternating Current Static Watt-hour Meters for Active Energy (classes 1 and 2) (GB/17315-2002/IEC 61036:2000). This report adopts the 2S as the meter reading accuracy for conservative calculation.

The major monitored parameters are listed in table below.

Data/Parameters:	EG _{total output, y}
Data Unit:	MWh
Description	Total electricity supplied to the grid during year y.
Source of data to be used:	Electricity meter reading at the 110kv substation built by the project activity.
Description of measurement methods and procedures to be applied:	The parameter will be continuously measured by the on-site bidirectional meter and monthly recorded. Data will be archived for 2 years following the end of the crediting period.
QA/QC procedures to be applied:	The accuracy of the meter is 0.5s, and the calibration frequency is once a year following the relevant national standard (DL/T 448-2000). Receipts from electricity sales will be used for cross check.

Data/Parameters:	EG _{total import, y}
Data Unit:	MWh
Description	Total electricity imported from the grid during year y.
Source of data to be used:	Electricity meter reading at the 110kv substation built by the project activity.
Description of measurement methods and procedures to be applied:	The readings of the electricity meter are continuously measured and monthly recorded.
QA/QC procedures to be applied:	The accuracy of the meter is 0.5s, and the calibration frequency is once a year following the relevant national standard (DL/T 448-2000). Receipts from electricity sales will be used for cross check.

Data/Parameters:	$EG_{\text{new output, y}}$
Data Unit:	MWh
Description	Total electricity generated by 8 new wind turbines during year y.
Source of data to be used:	Electricity meter reading inside of the turbine.
Description of measurement methods and procedures to be applied:	The parameter will be continuously measured by the power meter and daily recorded. Data will be archived for 2 years following the end of the crediting period.
QA/QC procedures to be applied:	The readings from the power meters inside of 8 new turbines will be recorded by on-site staff in daily logs everyday at 20:00. The data is also continuously measured by the central computer monitoring system at the project substation and will be used for cross checking the daily log.

Figure 3-1 shows a diagram for the metering points of the proposed project, and Table 3-2 lists details of all the meters.

Figure 3.1 Project meter diagram

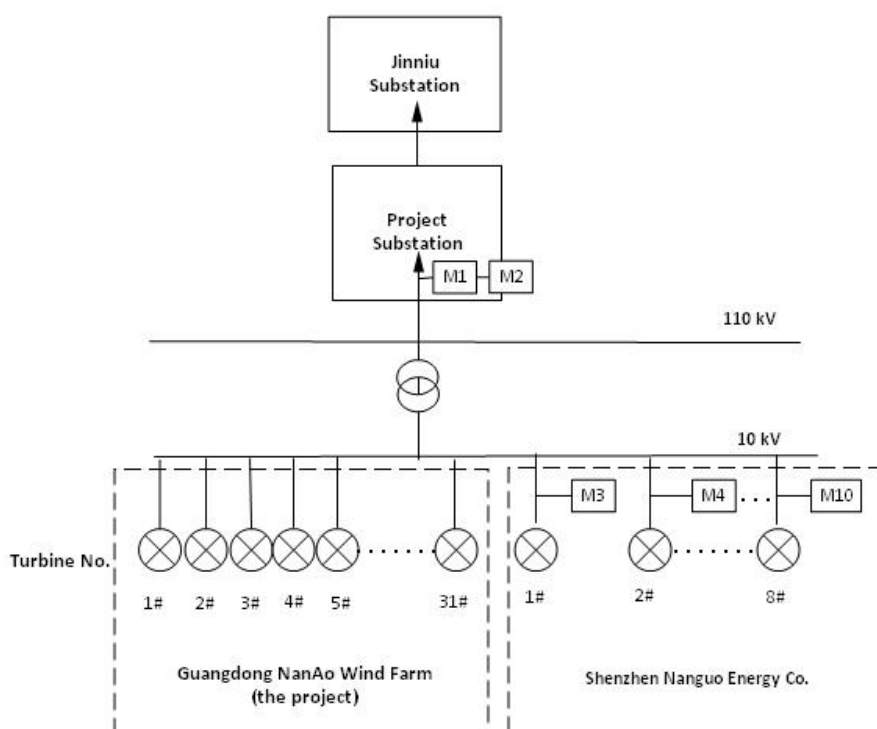


Table 3-2 Meter details

Name	Data Monitored	Recorded by	Accuracy	Calibration Date	Calibration Frequency
M1	$EG_{\text{total output, y}}$ $EG_{\text{total import, y}}$	Power grid company	0.5S	16 Jan. 2008	Once a year
M2	$EG_{\text{total output, y}}$	Power grid	0.5S	16 Jan. 2008	Once a year

(backup meter of M1)	EG _{total import, y}	company			
M3	Electricity generated by new turbine 1# (C7)	Project owner	2 S	9 Oct. 2008	Once a year
M4	Electricity generated by new turbine 2# (C8)		2 S	11 Nov. 2008	Once a year
M5	Electricity generated by new turbine 3# (C9)		2 S	21 Oct. 2008	Once a year
M6	Electricity generated by new turbine 4# (C10)		2 S	22 Oct. 2008	Once a year
M7	Electricity generated by new turbine 5# (C11)		2 S	27 Oct. 2008	Once a year
M8	Electricity generated by new turbine 6# (D9)		2 S	30 Oct. 2008	Once a year
M9	Electricity generated by new turbine 7# (D10)		2 S	5 Oct. 2008	Once a year
M10	Electricity generated by new turbine 8# (D11)		2 S	14 Oct. 2008	Once a year

3.3 QA/QC procedure

1) Project main meter and its backup meter were installed by certified technicians from local grid company, and are calibrated in accordance with the relevant national and industrial regulations (DL/T 448-2000) at least once a year by a qualified calibration entity. Eight PM 810 meters inside of each new wind turbine were installed by the manufacturer.

2) The total output and input electricity generations are continuously monitored by the main meter and recorded on 27th or 28th of each month at 20:00. The electricity generations of 8 new turbines are recorded by on-site staff every day at 20:00. The terminal computer system also records the power generation from each turbine of the project activity and of 8 new turbines.

3) All data and records of meter readings, meter calibration, and other documentation will be archived in both electronic and written forms, and be kept by the project owner for 2 years after the end of the last crediting period.

4) In case the main meter operates abnormally, the readings from the backup meter will be used. No abnormal operation has occurred during the VCU crediting period. .

5) Training on CDM monitoring was provided to the relevant staff on 26 August 2008 to guarantee the success of the implementation of the monitoring plan.

4. GHG Calculations

4.1 Project activity emissions

The proposed project is a zero-emission electricity generating activity. Therefore no emission from the project activity were identified, and $PE_y = 0$.

4.2 Leakages

According to the methodology ACM0002 (Version 6), the project leakage is not considered, and therefore, $L_y = 0$.

4.3 Baseline emissions

Baseline Emissions are calculated by multiplying the baseline emission factor by annual power generation.

$$BE_y = EG_y \times EF_y$$

Where:

BE_y is Baseline emissions of China Southern Power Grid in year y

EG_y is Net electricity supplied by the project to the grid in year y

EF_y is Emission factor in year y (calculated ex-ante and will not be changed during this monitoring period)

$$EG_y = EG_{\text{output}, y} - EG_{\text{import}, y} - EG_{\text{new output}, y}$$

Where:

$EG_{\text{output}, y}$ is total electricity supplied to the grid during year y;

$EG_{\text{import}, y}$ is total electricity imported from the grid during year y;

$EG_{\text{new output}, y}$ is total electricity generated by 8 new wind turbines during year y.

As a conservative approach to calculate the project emission reductions, the project will take the shared line losses of the Shenzhen Nanguo Energy Company. In addition, since the accuracy of the PM810 is 2S during the VCU crediting period, which is lower than that of main meter with a the accuracy of 0.5S. To be conservative, the current monitored data from new 8 turbines will be calculated as:

$$\text{Modified electricity generation of 8 new turbines} = EG_{\text{new output}, y} \times (1+2\%)$$

In addition, since the power output of 8 new turbines are recorded at 20:00 on each day, to be conservative, the total electricity generated by 8 new turbines will be calculated till 15 Dec. 20:00.

Emission reductions

The project emission reductions(ER_y) are calculated as the baseline emissions minus the project emissions.

$$\begin{aligned} ER_y &= BE_y - PE_y - LE_y \\ &= EG_{\text{output}, y} - EG_{\text{import}, y} - EG_{\text{new output}, y} - 0 - 0 \\ &= EG_{\text{total}, y} - EG_{\text{new output}, y} \end{aligned}$$

An emission factor (EF_y) of 0.9276 tCO₂e/MWh is used for the calculation, please refer to project CDM PDD for more details of calculation.

The result of the calculation is summarized in the following Table4-1 and Table 4-2:

Table 4-1 Emission reductions calculation of Guandong Nan Ao 26MW Wind Power Project

Period		Net electricity to the grid ($EG_{\text{total}, y}$)	Baseline emission (BE_y)	Project emission (PE_y)	Emission reductions (ER_y)
		MWh	tCO ₂ e	tCO ₂ e	tCO ₂ e
			$EG_y \times EF_y$		$BE_y - PE_y$
2008	1 Jan.- 27 Feb.*	10,136.533	9,402.65	0	9,402.65
	27 Feb. - 28 Mar.	5,483.241	5,086.25	0	5,086.25

28 Mar. - 28 Apr.	4,945.786	4,587.71	0	4,587.71
28 Apr. - 28 May	4,613.341	4,279.34	0	4,279.34
28 May - 28 Jun.	2,993.629	2,776.89	0	2,776.89
28 Jun. - 28 Jul.	1,435.722	1,331.78	0	1,331.78
28 Jul. - 28 Aug.	2,612.227	2,423.10	0	2,423.10
28 Aug. - 27 Sept.	2,103.045	1,950.78	0	1,950.78
27 Sept. - 28 Oct.	7,549.111	7,002.56	0	7,002.56
28 Oct. - 28 Nov.	8,787.628	8,151.40	0	8,151.40
28 Nov. 20:00 - 15 Dec. 0:00	4,734.400	4,391.63	0	4,391.63
Total	55,394.663	51,384.089	0	51,384.09
Total electricity generation of 8 new wind turbines by 20:00 15 Dec 2008 (EG_{new, output}) X (1+2%)*	3,083.095	2,860	0	2,860
Electricity generated by the project activity	52,311.568	48,524.211	0.000	48,524.211

* For conservative estimation, the total electricity of 8 new turbines were calculated until 20:00 of 15 Dec. when the daily log is recorded and is multiplied with a coefficient of (1+2%) to adjust the meter accuracy as described before.

Table 4-2 Electricity generated by 8 new wind turbines (Oct.-Dec. 2008)

Electricity generated by 8 new wind turbines (MWh)	
5 - 31 Oct. 2008	507.322
1 - 30 Nov.2008	1,767.972
1 - 15 Dec. 2008	747.348
Total	3,022.642

Therefore, the total emissions reduction generated during the project VCU crediting period is shown in the table below.

Monitoring Period	Emission reduction (tCO₂e)
01/01/2008~14/12/2008	48,524